

Assignment 1: Probability & Random Processes

(1) Bayes' theorem

Problem: A telegraphic communications system transmits dots and dashes. Assume that due to noise, an average of 2/5 of the dots and 1/3 of the dashes are changed. Suppose that the ratio between the transmitted dots and the transmitted dashes is 5 : 3. (a) What is the probability that a received signal will be the same as the transmitted signal if (i) the received signal is a dot? (ii) the received signal is a dash? (b) Assuming independence between signals, if the message dot-dot was received, what is the probability distribution of the four possible messages that could have been sent?

Solution: (a) Let A be the event that a dot is received, and B that a dash is received. Hypotheses: H_1 (dot sent), H_2 (dash sent). $P(H_1) = 5/8$, $P(H_2) = 3/8$. Conditional probabilities: $P(A|H_1) = 3/5$, $P(B|H_1) = 2/5$, $P(A|H_2) = 1/3$, $P(B|H_2) = 2/3$. Total probabilities: $P(A) = \frac{5}{8} \cdot \frac{3}{5} + \frac{3}{8} \cdot \frac{1}{3} = \frac{1}{2}$, $P(B) = \frac{5}{8} \cdot \frac{2}{5} + \frac{3}{8} \cdot \frac{2}{3} = \frac{1}{2}$ (i) $P(H_1|A) = \frac{P(H_1)P(A|H_1)}{P(A)} = \frac{(5/8) \times (3/5)}{1/2} = \frac{3}{4}$ (ii) $P(H_2|B) = \frac{P(H_2)P(B|H_2)}{P(B)} = \frac{(3/8) \times (2/3)}{1/2} = \frac{1}{2}$

(b) Distribution of four possibilities (A happened twice): H_2 - H_2 (dash-dash): $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$ H_2 - H_1 (dash-dot): $\frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$ H_1 - H_2 (dot-dash): $\frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$ H_1 - H_1 (dot-dot): $\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

(2) Characteristic function

Problem: A random variable X has PDF $f(x) = \frac{1}{2}e^{-|x|}$. (a) Find its characteristic function. (b) Find its mean and variance from it.

Solution: (a) Let $M(u)$ be characteristic function:

$$\begin{aligned} M(u) &= \int_{-\infty}^{+\infty} e^{iux} \frac{1}{2} e^{-|x|} dx \\ &= \frac{1}{2} \int_0^{\infty} e^{(iu-1)x} dx + \frac{1}{2} \int_{-\infty}^0 e^{(iu+1)x} dx \\ &= \frac{1}{2} \left(\frac{1}{1-iu} + \frac{1}{1+iu} \right) = \frac{1}{1+u^2} \end{aligned}$$

(b) Mean and Variance from Characteristic Function:
By definition, the characteristic function is an expectation:

$$M(u) = E[e^{iuX}] = \int_{-\infty}^{\infty} e^{iuX} f(x) dx$$

Taking the n -th derivative with respect to u brings down $(ix)^n$ from the exponent. Evaluating at $u = 0$ sets $e^{i(0)x} = 1$, leaving the integral of $x^n f(x)$. This gives the moment formula:

$$\langle x^n \rangle = (-i)^n M^{(n)}(0)$$

Using $M(u) = (1+u^2)^{-1}$, we can find the mean and variance:

$$\text{Mean: } \langle x \rangle = -iM'(0) = -i(0) = 0$$

$$\text{Variance: } \langle x^2 \rangle - \langle x \rangle^2 = (-i)^2 M''(0) - 0 = (-1)(-2) = 2$$

(3) Autocorrelation

Problem: Let $X(t) \sim N(0, \sigma^2)$. $Z(t) = \frac{1}{2} \left[1 + \frac{X(t)X(t+\tau)}{|X(t)X(t+\tau)|} \right]$. Show time average $\bar{Z}(\tau) = \frac{1}{\pi} \arccos[-\rho(\tau)]$.

Solution: Let $x = X(t)$, $y = X(t + \tau)$. Joint distribution: $f(x, y) = \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} \exp\left(-\frac{x^2+y^2-2\rho xy}{2(1-\rho^2)\sigma^2}\right)$ Since $\frac{1}{2} \left[1 + \frac{xy}{|xy|} \right]$ is 0 if opposite signs and 1 if same sign: $\bar{Z} = \int_0^0 \int_0^0 f(x, y) dx dy + \int_0^{\infty} \int_0^{\infty} f(x, y) dx dy = 2 \int_0^{\infty} \int_0^{\infty} f(x, y) dx dy$

Change to polar: $x = r \cos \theta$, $y = r \sin \theta$. Let $z^2 = \frac{r^2(1-\rho \sin 2\theta)}{2(1-\rho^2)\sigma^2}$.

$$\begin{aligned} \bar{Z} &= \frac{2}{2\pi\sigma^2\sqrt{1-\rho^2}} \int_0^{\frac{\pi}{2}} \int_0^{\infty} e^{-z^2} r dr d\theta \\ &= \frac{\sqrt{1-\rho^2}}{\pi} \int_0^{\frac{\pi}{2}} \frac{d\theta}{1-\rho \sin 2\theta} = \frac{1}{\pi} \arctan \left(\frac{\tan \theta - \rho}{\sqrt{1-\rho^2}} \right) \Big|_0^{\frac{\pi}{2}} \\ &= \frac{1}{\pi} \left[\frac{\pi}{2} - \arctan \left(\frac{-\rho}{\sqrt{1-\rho^2}} \right) \right] \end{aligned}$$

Using a right-angled triangle, this equals $\frac{1}{\pi} \arccos[-\rho(\tau)]$.

(4) Radar system

Problem: A radar system uses radio waves to detect aircraft. Let X be the received signal: $X = W$ (if no aircraft), and $X = 1 + W$ (if aircraft present), where noise $W \sim N(0, 1/9)$. We can write $X = \theta + W$. (a) Write H_0 and H_1 in terms of θ . (b) Design a test with significance level $\alpha = 0.05$. (c) Find the probability β of Type II error. (d) If $X = 0.6$, reject H_0 at $\alpha = 0.01$? (e) If we want missing probability $< 5\%$, what is the smallest α ? (f) Design a test using the likelihood ratio $\Lambda(X)$ with threshold η .

Solution: Note: Since $W \sim N(0, \sigma^2 = 1/9)$, its standard deviation is $\sigma = 1/3$. The standard normal variable is $Z = \frac{W}{1/3} = 3W \sim N(0, 1)$.

(a) **Hypotheses Formulation** $H_0 : \theta = 0$ (Null hypothesis: no aircraft is present). $H_1 : \theta = 1$ (Alternative hypothesis: an aircraft is present).

(b) **Decision Rule** for $\alpha = 0.05$ We set a threshold c . We reject H_0 if $X > c$. The significance level α (Type I error) is:

$$\begin{aligned} \alpha &= P(\text{Reject } H_0 \mid H_0) = P(X > c \mid \theta = 0) \\ &= P(W > c) = P(3W > 3c) = P(Z > 3c) \end{aligned}$$

Using the standard normal CDF Φ , $P(Z > 3c) = 1 - \Phi(3c)$. We want $1 - \Phi(3c) = 0.05 \implies \Phi(3c) = 0.95$. From the Z-table, the value is ≈ 1.645 . $3c = 1.645 \implies c \approx 0.548$. *Test design:* Reject H_0 if $X > 0.548$.

(c) **Type II Error** β Type II error occurs when we accept H_0 but H_1 is true:

$$\begin{aligned} \beta &= P(\text{Accept } H_0 \mid H_1) = P(X < c \mid \theta = 1) \\ &= P(1 + W < c) = P(W < c - 1) \\ &= P(3W < 3(c - 1)) = \Phi(3(c - 1)) \end{aligned}$$

Substitute $c = 0.548$: $\beta = \Phi(3(-0.452)) = \Phi(-1.356) \approx 0.087$.

(d) **Evaluating at** $\alpha = 0.01$ For $\alpha = 0.01$, we need a new threshold c' : $1 - \Phi(3c') = 0.01 \implies \Phi(3c') = 0.99$. From the Z-table, $3c' \approx 2.33 \implies c' \approx 0.776$. Since the observed $X = 0.6$ is less than 0.776, we **cannot** reject H_0 at the 0.01 level.

(e) **Minimizing Type II Error** We want $\beta \leq 0.05$. From part (c), $\beta = \Phi(3(c - 1))$. $\Phi(3(c - 1)) \leq 0.05 \implies 3(c - 1) \leq -1.645 \implies c \leq 0.452$. Using $c = 0.452$, the smallest corresponding α is: $\alpha \geq 1 - \Phi(3 \times 0.452) = 1 - \Phi(1.356) \approx 0.087$. Thus, the smallest significance level we can achieve is ≈ 0.087 .

(f) **Likelihood Ratio Test** The PDFs for X under H_0 and H_1 are: $f(x \mid H_0) = \frac{3}{\sqrt{2\pi}} e^{-\frac{9x^2}{2}}$ $f(x \mid H_1) = \frac{3}{\sqrt{2\pi}} e^{-\frac{9(x-1)^2}{2}}$ The likelihood ratio $\Lambda(x)$ is:

$$\begin{aligned} \Lambda(x) &= \frac{f(x \mid H_1)}{f(x \mid H_0)} = \frac{e^{-\frac{9}{2}(x^2 - 2x + 1)}}{e^{-\frac{9}{2}x^2}} \\ &= e^{-\frac{9}{2}(-2x+1)} = e^{\frac{9(2x-1)}{2}} \end{aligned}$$

We accept H_1 if $\Lambda(X) > \eta$, which implies: $e^{\frac{9(2x-1)}{2}} > \eta \implies \frac{9(2x-1)}{2} > \ln \eta \implies X > \frac{1}{2} \left(1 + \frac{2}{9} \ln \eta \right)$.

Assignment 2: Information Theory Basics

(1) Average entropy

Problem: Let $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ be the entropy function of a binary source. (a) Evaluate $H(1/3)$. (b) Calculate the expected entropy $H(p)$ when p is chosen uniformly in $[0, 1]$.

Solution: (a) $H(\frac{1}{3}) = -\frac{1}{3} \log_2 \frac{1}{3} - (1 - \frac{1}{3}) \log_2 (1 - \frac{1}{3}) = \log_2 3 - \frac{2}{3} = 0.918$ bits. (b) The expected entropy is:

$$\begin{aligned} H(p) &= \frac{\int_0^1 [-p \log_2 p - (1-p) \log_2 (1-p)] dp}{1-0} \\ &= - \int_0^1 \frac{p \ln p + (1-p) \ln(1-p)}{\ln 2} dp = - \frac{2}{\ln 2} \int_0^1 x \ln x dx \\ &= - \frac{2}{\ln 2} \left[\frac{x^2}{2} \ln x - \frac{x^2}{4} \right]_0^1 = \frac{1}{2 \ln 2} = 0.721 \text{ bits.} \end{aligned}$$

(2) Mutual info for correlated normal distributions

Problem: $X, Y \sim N_2(0, \Sigma)$ with $\Sigma = \begin{pmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{pmatrix}$. Evaluate $I(X; Y)$ and comment on $\rho = 1, 0, -1$.

Solution: Marginal entropy of $X \sim N(0, \sigma^2)$:

$$\begin{aligned} H(X) &= - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \left[-\frac{x^2}{2\sigma^2} - \ln \sqrt{2\pi\sigma^2} \right] dx \\ &= \frac{1}{2} + \frac{1}{2} \ln(2\pi\sigma^2) \end{aligned}$$

Joint entropy $H(X, Y) = - \iint f(x, y) \ln f(x, y) dx dy$. Substitute $u = \frac{x - \rho y}{\sqrt{1 - \rho^2}}$ and $v = y$ (so $x = u\sqrt{1 - \rho^2} + v\rho$). The Jacobian determinant is $\sqrt{1 - \rho^2}$. The integral splits into two parts. The quadratic part evaluates to:

$$\begin{aligned} &\iint \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}} \left(\frac{u^2+v^2}{2\sigma^2} \right) du dv \\ &= \frac{1}{2\pi\sigma^2} \int_0^{2\pi} \int_0^{\infty} e^{-\frac{r^2}{2\sigma^2}} \frac{r^2}{2\sigma^2} r dr d\theta = 1 \end{aligned}$$

The logarithmic constant part evaluates to $\ln(2\pi\sigma^2\sqrt{1-\rho^2})$. Thus, $H(X, Y) = 1 + \ln(2\pi\sigma^2\sqrt{1-\rho^2})$. Mutual information $I(X; Y) = H(X) + H(Y) - H(X, Y)$:

$$\begin{aligned} I(X; Y) &= \left[\frac{1}{2} + \frac{1}{2} \ln(2\pi\sigma^2) \right] + \left[\frac{1}{2} + \frac{1}{2} \ln(2\pi\sigma^2) \right] \\ &\quad - \left[1 + \ln(2\pi\sigma^2\sqrt{1-\rho^2}) \right] = -\frac{1}{2} \ln(1 - \rho^2) \end{aligned}$$

Comments: $-\rho = 1$: $I \rightarrow \infty$ (knowing X implies perfect knowledge of Y). $-\rho = 0$: $I = 0$ (X and Y are independent). $-\rho = -1$: $I \rightarrow \infty$ (knowing X implies perfect knowledge of Y).

(3) Channel capacity

Problem: Binary asymmetric channel. Input $X = \{0, 1\}$ with probs $\{0.5, 0.5\}$. Matrix $\begin{pmatrix} 1-\alpha & \beta \\ \alpha & 1-\beta \end{pmatrix}$. Find $H(X)$, $p(Y)$, $H(Y)$, $p(X, Y)$, $H(X, Y)$, $I(X; Y)$, and max/min capacity conditions.

Solution: (a) Input Entropy:

$$H(X) = -p \log_2 p - (1-p) \log_2 (1-p)$$

$$H(X) = -0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 1 \text{ bit.}$$

(b) Output Distribution & Entropy: By total probability, the outputs are:

$$\begin{aligned} p(y=0) &= 0.5(1-\alpha) + 0.5\beta = 0.5(1-\alpha+\beta) \\ p(y=1) &= 0.5\alpha + 0.5(1-\beta) = 0.5(\alpha+1-\beta) \end{aligned}$$

The output entropy $H(Y) = -\sum p(y) \log_2 p(y)$. Expanding and extracting the 1/2 gives a leading +1:

$$\begin{aligned} H(Y) &= 1 - 0.5(1-\alpha+\beta) \log_2 (1-\alpha+\beta) \\ &\quad - 0.5(\alpha+1-\beta) \log_2 (\alpha+1-\beta) \end{aligned}$$

(c) Joint Distribution & Entropy: The joint probabilities $p(x, y) = p(x)p(y|x)$ form: $p(X, Y) = \begin{pmatrix} 0.5(1-\alpha) & 0.5\beta \\ 0.5\alpha & 0.5(1-\beta) \end{pmatrix}$ The joint entropy $H(X, Y) = -\sum p(x, y) \log_2 p(x, y)$. Expanding and extracting the 1/2 gives:

$$\begin{aligned} H(X, Y) &= 1 - 0.5\alpha \log_2 \alpha - 0.5(1-\alpha) \log_2 (1-\alpha) \\ &\quad - 0.5\beta \log_2 \beta - 0.5(1-\beta) \log_2 (1-\beta) \end{aligned}$$

(d) Mutual Information: $I(X; Y) = H(X) + H(Y) - H(X, Y)$.**(e) Maximum Capacity:** $I(X; Y)$ reaches its maximum (1 bit) when $(\alpha, \beta) = (0, 0)$ or $(1, 1)$. (Perfect transmission or perfectly inverted).**(f) Minimum Capacity:** $I(X; Y)$ reaches its minimum (0 bits) when $\alpha + \beta = 1$. (Output is completely independent of the input).**(4) Shannon, Fano, and Huffman codes****Problem:** $X = \{a, b, c, d, e, f, g\}$ with probs $\{0.01, 0.24, 0.05, 0.20, 0.47, 0.01, 0.02\}$. Find $H(X)$, codewords, avg lengths, and optimal codes.**Solution: (a) Entropy of X:** $H(X) = -\sum p_i \log_2 p_i$ $H(X) = -(0.01 \log_2 0.01 + 0.24 \log_2 0.24 + \dots + 0.02 \log_2 0.02) = 1.932$ bits.**(b) Codeword Generation:** (Note: Sort probabilities in descending order first: e:0.47, b:0.24, d:0.20, c:0.05, g:0.02, a:0.01, f:0.01)**Shannon Code:** Length $l_i = \lceil -\log_2 p_i \rceil$. Cumulative probability $F_i = \sum_{k=1}^{i-1} p_k$. Codeword is the first l_i bits of the binary expansion of F_i : a:1000110, b:010, c:10000, d:011, e:00, f:1000111, g:100010.**Fano (tree split):** a:111110, b:10, c:1110, d:110, e:0, f:111111, g:11110.**Huffman (bottom-up grouping):** a:000000, b:01, c:0001, d:001, e:1, f:000001, g:00001.**(c) Expected lengths:** $L = \sum p_i l_i$

Shannon = 2.77 bits. Fano = 1.97 bits. Huffman = 1.97 bits.

(d) Both Fano and Huffman are optimal. Greater than $H(X)$ by $1.970 - 1.932 = 0.038$ bits.**Assignment 3: Adv. Entropy****(1) Conditioning reduces entropy****Problem:** Verify if $H(X|Y=y)$ can be $>$ or $<$ $H(X)$ using a given joint distribution table. Compute all marginal, joint, conditional entropies, and mutual information.**Solution: (a) Marginal probabilities $P(X)$ and $P(Y)$:** $P(X) = \{1/2, 1/4, 1/8, 1/8\}$ $P(Y) = \{1/4, 1/4, 1/4, 1/4\}$ **(b) Joint entropy $H(X, Y)$:** $H(X, Y) = -\sum_{x,y} p(x, y) \log_2 p(x, y) = \frac{1}{4} \log_2 4 + \frac{1}{4} \log_2 8 + \frac{3}{8} \log_2 16 + \frac{1}{8} \log_2 32 = \frac{27}{8}$ bits.**(c) Marginal entropies $H(X)$ and $H(Y)$:** $H(X) = \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{2}{8} \log_2 8 = \frac{7}{4}$ bits. $H(Y) = 4 \times \frac{1}{4} \log_2 4 = 2$ bits.**(d) Conditional entropy $H(X|Y=y)$ for each y :** (Note: Redraw the conditional probability table $p(x|y) = P(X=x|Y=y)$ first!) $H(X|Y=1) = \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{2}{8} \log_2 8 = \frac{7}{4}$ bits. $H(X|Y=2) = \frac{1}{4} \log_2 4 + \frac{1}{2} \log_2 2 + \frac{3}{8} \log_2 8 = \frac{7}{4}$ bits. $H(X|Y=3) = 4 \times \frac{1}{4} \log_2 4 = 2$ bits. $H(X|Y=4) = 1 \log_2 1 = 0$ bits.**(e) Comparison with $H(X)$:** (i) $H(X|Y=y) < H(X)$: This happens when $y = 4$ (since $0 < 7/4$). (ii) $H(X|Y=y) > H(X)$: This happens when $y = 3$ (since $2 > 7/4$).**(f) Conditional entropies $H(X|Y)$ and $H(Y|X)$:** $H(X|Y) = \sum_y P(Y=y) H(X|Y=y) = \frac{1}{4} [\frac{7}{4} + \frac{7}{4} + 2 + 0] = \frac{13}{8}$ bits. $H(Y|X) = \frac{13}{8}$ bits.**(g) Mutual information $I(X; Y)$:** $I(X; Y) = H(X) - H(X|Y) = \frac{7}{4} - \frac{11}{8} = \frac{3}{8}$ bits.**(2) Principle of maximum entropy****Problem:** (a) Find the best guess for the distribution of a 6-face dice using the principle of maximum entropy if: (i) no other info is given, (ii) its average is known to be 25/6. (b) For a gas system with fixed temperature T , show that the Maxwell-Boltzmann distribution maximizes the Shannon entropy.**Solution: (a) Six-face dice distribution:** (i) *No information:* By the principle of maximum entropy, all outcomes are equally likely. Thus, uniform distribution: $p_i = 1/6$. (ii) *Known average (25/6):* We maximize entropy subject to two constraints: $\sum p_i = 1$ and $\sum ip_i = 25/6$. Set up the Lagrangian $J(p)$: $J(p) = -\sum_{i=1}^6 p_i \ln p_i + \lambda_0 (\sum_{i=1}^6 p_i - 1) + \lambda_1 (\sum_{i=1}^6 ip_i - \frac{25}{6})$ Take the partial derivative w.r.t p_i and set to zero: $\frac{\partial J}{\partial p_i} = -1 - \ln p_i + \lambda_0 + \lambda_1 i = 0 \Rightarrow p_i = e^{-1+\lambda_0+\lambda_1 i}$ (Note: Solving this numerically gives $\lambda_1 \approx 0.236$ and $p_i = \{0.085, 0.108, 0.137, 0.173, 0.219, 0.278\}$).**(b) Gas system (Maxwell-Boltzmann):** We maximize $H = -\sum p_i \ln p_i$ subject to $\sum p_i = 1$ and mean energy $\sum p_i \epsilon_i = \bar{\epsilon}$. Set up the Lagrangian $J(p)$: $J(p) = -\sum_{i=1}^n p_i \ln p_i + \lambda_0 (\sum_{i=1}^n p_i - 1) + \lambda_1 (\sum_{i=1}^n p_i \epsilon_i - \bar{\epsilon})$ Differentiating w.r.t p_i and setting to zero gives: $\frac{\partial J}{\partial p_i} = -1 - \ln p_i + \lambda_0 + \lambda_1 \epsilon_i = 0 \Rightarrow p_i = e^{\lambda_1 \epsilon_i - 1 + \lambda_0} = \frac{e^{\lambda_1 \epsilon_i}}{e^{1-\lambda_0}}$ To eliminate the constant λ_0 , we use the probability constraint $\sum p_i = 1$: $\sum_{i=1}^n \frac{e^{\lambda_1 \epsilon_i}}{e^{1-\lambda_0}} = 1 \Rightarrow e^{1-\lambda_0} = \sum_{i=1}^n e^{\lambda_1 \epsilon_i}$ Substitute $e^{1-\lambda_0}$ back into the p_i equation: $p_i = \frac{e^{\lambda_1 \epsilon_i}}{\sum_{i=1}^n e^{\lambda_1 \epsilon_i}}$ By substituting the physics constant $\lambda_1 = -1/(kT)$, we obtain the exact Maxwell-Boltzmann distribution.**(3) Differential entropy****Problem:** Calculate the differential entropy $H(X) = -\int f(x) \ln f(x) dx$ for: (a) Uniform, (b) Normal, (c) Exponential, and (d) Laplace distributions.**Solution: (a) Uniform distribution on $[0, a]$:** $f(x) = \frac{1}{a}$ for $0 \leq x \leq a$. $H(X) = -\int_0^a \frac{1}{a} \ln \left(\frac{1}{a} \right) dx = -\frac{1}{a} (-\ln a) \times [x]_0^a = \ln a$. Note: If $a < 1$, then $\ln a < 0$, meaning $H(X)$ is negative. Unlike discrete entropy (which is always ≥ 0), differential entropy can be negative!**(b) Normal distribution $N(0, \sigma^2)$:** (Derived using the results from Assignment 2): $H(X) = \frac{1}{2} + \frac{1}{2} \ln(2\pi\sigma^2)$ **(c) Exponential distribution $f(x) = \lambda e^{-\lambda x}$ ($x \geq 0$):** Instead of complex integration by parts, we can split the integral using the properties of the PDF (total probability is 1, and mean is $1/\lambda$):

$$\begin{aligned} H(X) &= -\int_0^\infty \lambda e^{-\lambda x} (\ln \lambda - \lambda x) dx \\ &= -\ln \lambda \underbrace{\int_0^\infty \lambda e^{-\lambda x} dx}_{\text{Total Probability} = 1} + \lambda \underbrace{\int_0^\infty x (\lambda e^{-\lambda x}) dx}_{\text{Mean of Exp} = 1/\lambda} \\ &= -\ln \lambda (1) + \lambda \left(\frac{1}{\lambda} \right) = 1 - \ln \lambda \end{aligned}$$

(d) Laplace distribution $f(x) = \frac{\lambda}{2} e^{-\lambda|x|}$: Because the function is symmetric around $x = 0$, the integral over $(-\infty, \infty)$ is exactly twice the integral over $(0, \infty)$:

$$\begin{aligned} H(X) &= -\int_{-\infty}^\infty \frac{\lambda}{2} e^{-\lambda|x|} \left(\ln \frac{\lambda}{2} - \lambda|x| \right) dx \\ &= -2 \int_0^\infty \frac{\lambda}{2} e^{-\lambda x} \left(\ln \frac{\lambda}{2} - \lambda x \right) dx \\ &= -\ln \left(\frac{\lambda}{2} \right) \underbrace{\int_0^\infty \lambda e^{-\lambda x} dx}_{=1} + \underbrace{\int_0^\infty \lambda x (\lambda e^{-\lambda x}) dx}_{=1/\lambda} \\ &= -\ln \left(\frac{\lambda}{2} \right) (1) + 1 = 1 - \ln \left(\frac{\lambda}{2} \right) \end{aligned}$$

(4) Relative entropy**Problem:** Prove the divergence inequality $D(p||q) \geq 0$ using a mathematical function. Examine its symmetry for a binary channel.**Solution: (a) Mathematical Foundation (Function Maximization):** Consider the function $f(a) = \log a - a + 1$. Take the first and second derivatives: $f'(a) = \frac{1}{a} - 1$ and $f''(a) = -\frac{1}{a^2}$. Setting $f'(a) = 0$ gives a critical point at $a = 1$. Since the second derivative $f''(a) < 0$ for all $a > 0$, the function attains its global maximum at $a = 1$. The maximum value is $f(1) = \log(1) - 1 + 1 = 0$. Since the maximum is 0, it must be true that $f(a) \leq 0$ for all $a > 0$. This directly implies the fundamental inequality: $-\log a \geq 1 - a$.**(b) Proof of Non-negativity $D(p||q) \geq 0$:** By definition, the relative entropy is: $D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \sum_x p(x) \left[-\log \frac{q(x)}{p(x)} \right]$ Using the inequality from part (a), we substitute $a = \frac{q(x)}{p(x)}$ into $-\log a \geq 1 - a$:

$$\begin{aligned} D(p||q) &\geq \sum_x p(x) \left[1 - \frac{q(x)}{p(x)} \right] \\ &= \sum_x p(x) - \sum_x p(x) \frac{q(x)}{p(x)} \\ &= \sum_x p(x) - \sum_x q(x) \end{aligned}$$

Since $p(x)$ and $q(x)$ are both valid probability distributions, the sum of all their probabilities must equal 1 (i.e., $\sum p(x) = 1$ and $\sum q(x) = 1$). Therefore, $D(p||q) \geq 1 - 1 = 0$. (Proof completed).**(c) Symmetry in a Binary Channel:** Usually, $D(p||q) \neq D(q||p)$, but there are exceptions. (i) Calculation for $p(0) = 0.3, q(0) = 0.7$:

$$\begin{aligned} D(p||q) &= 0.3 \log_2 \left(\frac{0.3}{0.7} \right) + 0.7 \log_2 \left(\frac{0.7}{0.3} \right) \approx 0.489 \text{ bits.} \\ D(q||p) &= 0.7 \log_2 \left(\frac{0.7}{0.3} \right) + 0.3 \log_2 \left(\frac{0.3}{0.7} \right) \approx 0.489 \text{ bits.} \end{aligned}$$

In this specific case, $D(p||q) = D(q||p)$.(ii) *Conditions for Symmetry:* For a binary channel, expanding the equation $D(p||q) = D(q||p)$ gives: $p(0) \log \frac{p(0)}{q(0)} + [1-p(0)] \log \frac{1-p(0)}{1-q(0)} = q(0) \log \frac{q(0)}{p(0)} + [1-q(0)] \log \frac{1-q(0)}{1-p(0)}$ This equality holds true if and only if $q(x) = p(x)$ (the distributions are completely identical) or $q(x) = 1 - p(x)$ (the probabilities are perfectly flipped, as demonstrated in part i).**Assignment 4: Time Series & Fourier****(1) Stationary ARMA Processes****Problem:** Given AR(2) process $(1 - 1.1B + 0.18B^2)Y_t = \varepsilon_t$ where B is the backward shift operator and ε_t is white noise with $E(\varepsilon_t \varepsilon_\tau) = \mathbf{1}[t = \tau]$. (a) Is it covariance-stationary? (b) If so, compute its autocovariance.**Solution: (a) Stationarity Test (Root Method):** Factor the AR polynomial: $(1 - 1.1c + 0.18c^2) = (1 - 0.9c)(1 - 0.2c)$ Roots are $c = 1/0.9 \approx 1.11$ and $c = 1/0.2 = 5$.**Condition:** An AR process is covariance-stationary if and only if **all roots lie strictly outside the unit circle** $|c| > 1$. Since both roots > 1 , **Yes, it is covariance-stationary.****(b) Autocovariance Calculation:** For AR(2): $Y_t = 1.1Y_{t-1} - 0.18Y_{t-2} + \varepsilon_t$. Multiply both sides by Y_{t-s} and take expectations to get the **Yule-Walker equations** $E(\varepsilon_t Y_t) = E(\varepsilon_t^2) = 1$; $E(\varepsilon_t Y_t - 1) = 0$; **From(3) with $s=2$:** $\gamma_2 = 1.1\gamma_1 - 0.18\gamma_0$ (4)

Substitute (4) into (1): $(1 - 0.18^2)\gamma_0 = 1.1(1 - 0.18)\gamma_1 + 1$
Solve (2) and (5) simultaneously:

$$\gamma_0 = 7.889, \quad \gamma_1 = 7.355, \quad \gamma_2 = 6.670$$

Higher lags: use recursion (3).

(2) Linear Regression & Unit Root Tests

Problem: You have a time series X_t with backward difference $\Delta X_t \equiv X_t - X_{t-1}$.

(a) You have first fitted the following model:

$$\Delta X_t = +0.51(\pm 0.29) - 0.11(\pm 0.04)X_{t-1}$$

$$-0.39(\pm 0.09)\Delta X_{t-1} - 0.32(\pm 0.09)\Delta X_{t-2} - 0.22(\pm 0.10)\Delta X_{t-3}$$

where the number after $\pm$ is the standard error. Now perform a unit root test.

(i) What is the t -statistic of X_{t-1} ? At what significance level can you reject H_0 ? (Use ADF table.)

(ii) What can you conclude if H_0 is rejected?

(b) After that, you have fitted three other models (Y is an independent variable):

$$\Delta X_t = +0.002(\pm 0.014) - 0.31(\pm 0.10)\Delta X_{t-1} \quad (1)$$

$$\Delta X_t = +0.021(\pm 0.158) - 0.46(\pm 0.10)\Delta X_{t-1} - 0.39(\pm 0.11)\Delta X_{t-2} - 0.25(\pm 0.08)\Delta X_{t-3} \quad (2)$$

$$\Delta X_t = +1.279(\pm 0.57) - 0.51(\pm 0.10)\Delta X_{t-1} - 0.44(\pm 0.11)\Delta X_{t-2} - 0.30(\pm 0.09)\Delta X_{t-3} - 0.16(\pm 0.07)Y_{t-1} \quad (3)$$

(i) What is the t -statistic of Y_{t-1} ? At what significance level can you reject H_0 for Y_{t-1} ? (Use a t -distribution table.)

(ii) Use models (1)–(3) and the following data to forecast X_5 :

t	X_t	Y_t
1	0.8	7.7
2	4.3	7.9
3	2.9	7.7
4	1.4	7.0

Solution:

(a)(i) **ADF t -statistic of X_{t-1} :** Model has a (Constant but NO Time Trend $\Rightarrow$ use τ_μ column in ADF table.

$$t = \frac{\hat{\beta}}{\text{SE}} = \frac{-0.11}{0.04} = -2.75$$

Critical values (τ_μ): -2.57 (10%), -2.86 (5%).
Since $-2.86 < -2.75 < -2.57$: **reject H_0 at 10% but not 5%.**

(a)(ii) **Conclusion upon rejecting H_0 :**

- ΔX_t depends on X_{t-1} (mean reversion)
- X_t does **not** have a unit root
- X_t is **stationary**

(b)(i) **t -statistic of Y_{t-1} in Model (3):**

$$t = \frac{-0.16}{0.07} \approx -2.29$$

Use two-sided t -table. Critical values: -1.96 (5%), -2.33 (2%).
Reject H_0 at 5% but not 2%: Y_{t-1} is a significant predictor.

(b)(ii) **First compute ΔX_t , then forecast:**

t	X_t	Y_t	ΔX_t
1	0.8	7.7	N/A
2	4.3	7.9	3.5
3	2.9	7.7	-1.4
4	1.4	7.0	-1.5

At $t = 5$: use $\Delta X_4 = -1.5$, $\Delta X_3 = -1.4$, $\Delta X_2 = 3.5$, $Y_4 = 7.0$.

Model (1):

$$\Delta X_5 = 0.002 - 0.31(-1.5) = 0.467 \Rightarrow X_5 = 1.4 + 0.467 = \mathbf{1.867}$$

Model (2):

$$\Delta X_5 = 0.021 - 0.46(-1.5) - 0.39(-1.4) - 0.25(3.5) = 0.382 \Rightarrow X_5 = \mathbf{1.782}$$

Model (3):

$$\Delta X_5 = 1.279 - 0.51(-1.5) - 0.44(-1.4) - 0.30(3.5) - 0.16(7.0) = 0.49 \Rightarrow X_5 = \mathbf{1.89}$$

(3) Fourier Transform of Impulses

Problem: In this question, you will learn about the Fourier Transform of impulses. Use the notations of Fourier Transform: $F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$.

(a) Let us begin by considering a Gaussian impulse

$$f(t) = ce^{-b(t-t_0)^2}$$

where t is the time variable and (b, c, t_0) are constants. What is its Fourier transform?

(b) A Dirac delta impulse $\delta(t - t_0)$ can be approximated by using the Gaussian impulse above and setting $b = 1/(2\sigma^2)$ and $c = 1/\sqrt{2\pi\sigma^2}$.

- (5) (i) A Dirac delta function $\delta(x)$ has the properties that $\delta(x) = \begin{cases} 0 & (x \neq 0) \\ \infty & (x = 0) \end{cases}$ and $\int_{-\infty}^{\infty} \delta(x) dx = 1$. Describe how the Gaussian impulse satisfies them as $\sigma \rightarrow 0$.
- (ii) Using this approximation and the result in (a), what is the Fourier transform of a Dirac delta impulse?
- (c) Next, we are going to consider some impulses that have temporal power law decay. Let us begin by considering a step function:

$$f(t) = \begin{cases} 1 & (t > 0) \\ 0 & (t < 0) \end{cases}$$

If you naively perform Fourier transform on it, you will get something that is undefined.

Instead, you can first perform Fourier transform on $f(t) = \begin{cases} e^{-\alpha t} & (t > 0) \\ 0 & (t < 0) \end{cases}$ and then let $\alpha \rightarrow 0$ to retrieve the Fourier transform of the step function.

- (i) Hence, what is the Fourier Transform of the step function? (Hint: its discontinuity at $t = 0$ will give a delta function in your answer.)
- (ii) What is its power spectral distribution function? Comment on your result.
- (d) Using the method shown in (c), what is the Fourier transform of the following sign function?

$$f(t) = \begin{cases} 1 & (t > 0) \\ -1 & (t < 0) \end{cases}$$

(e) Using the result in (d), what are the Fourier transforms of (i) $f(t) = 1/t$, (ii) $f(t) = |t|$, and (iii) $f(t) = 1/t^2$ for $-\infty \leq t \leq \infty$? [Hint: for $f(t) = 1/t$, start from its inverse Fourier transform and redefine the parameters in the integral, $t \rightarrow -\omega$ and $\omega \rightarrow t'$.]

Notation: $F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$

(a) **Gaussian Impulse** $f(t) = ce^{-b(t-t_0)^2}$:

The Fourier transform of the Gaussian impulse is

$$F(\omega) = \int_{-\infty}^{\infty} ce^{-b(t-t_0)^2} e^{-i\omega t} dt$$

For the exponent, we complete the square as follows.

$$b(t - t_0)^2 + i\omega t = b \left[t^2 - 2t_0 t + t_0^2 + \frac{i\omega t}{b} \right]$$

$$= b \left[\left(t - \left(t_0 - \frac{i\omega}{2b} \right) \right)^2 + \left(\frac{\omega}{2b} \right)^2 + \frac{it_0\omega}{b} \right]$$

Hence,

$$F(\omega) = ce^{-b \left[\left(\frac{\omega}{2b} \right)^2 + \frac{it_0\omega}{b} \right]} \int_{-\infty}^{\infty} e^{-b \left(t - \left(t_0 - \frac{i\omega}{2b} \right) \right)^2} dt$$

Define $t - \left(t_0 - \frac{i\omega}{2b} \right) = t'$, then recall $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ so

$$F(\omega) = c\sqrt{\frac{\pi}{b}} e^{-b \left[\left(\frac{\omega}{2b} \right)^2 + \frac{it_0\omega}{b} \right]}$$

(b) **Dirac Delta Impulse** $\delta(t - t_0)$:

Approximate with Gaussian: $b = 1/(2\sigma^2)$, $c = 1/\sqrt{2\pi\sigma^2}$.

i) The Gaussian impulse becomes

$$f(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(t-t_0)^2}$$

It is a Gaussian distribution with mean t_0 and variance σ^2 . Since it is normalized, $\int_{-\infty}^{\infty} f(t) dt = 1$ always holds.

As $\sigma \rightarrow 0$, the bell shape of $f(t)$ becomes so narrow that in order to maintain the unit area, its height must approach infinity. In other words, $f(t) = \begin{cases} 0 & (t \neq t_0) \\ \infty & (t = t_0) \end{cases}$.

(ii) The result in (a) becomes

$$F(\omega) = c\sqrt{\frac{\pi}{b}} e^{-b \left[\left(\frac{\omega}{2b} \right)^2 + \frac{it_0\omega}{b} \right]} = c\sqrt{\frac{\pi}{b}} e^{-\frac{\omega^2}{4b} - it_0\omega}$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \sqrt{\frac{\pi}{1/(2\sigma^2)}} e^{-\frac{\omega^2}{2/\sigma^2} - it_0\omega} = e^{-\frac{\sigma^2\omega^2}{2}} e^{-it_0\omega}$$

As $\sigma \rightarrow 0$, we get $F(\omega) = e^{-it_0\omega}$. Note that the Fourier Transform of the Dirac delta impulse has a magnitude 1 and a complex phase $t_0\omega$.

(c) **Step Function:** $f(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$

(i) The Fourier transform of the step function is

$$F(\omega) = \lim_{\alpha \rightarrow 0} \int_0^{\infty} e^{-\alpha t} e^{-i\omega t} dt$$

$$= \lim_{\alpha \rightarrow 0} \frac{1}{\alpha + i\omega} = \lim_{\alpha \rightarrow 0} \left(\frac{\alpha}{\alpha^2 + \omega^2} - \frac{i\omega}{\alpha^2 + \omega^2} \right)$$

To evaluate the limit, we must consider its real and imaginary parts separately. As $\alpha \rightarrow 0$,

$$\frac{\alpha}{\alpha^2 + \omega^2} \rightarrow \frac{0}{0 + \omega^2} \quad \text{and} \quad \frac{i\omega}{\alpha^2 + \omega^2} \rightarrow \frac{i}{\omega}$$

The real part becomes indeterminate if $\omega = 0$ but remains 0 otherwise. To model this behaviour, we first assume $\frac{\alpha}{\alpha^2 + \omega^2} = c\delta(\omega)$ for some constant c .

On one hand, we have

$$\lim_{\alpha \rightarrow 0} \int_{-\infty}^{+\infty} \frac{\alpha d\omega}{\alpha^2 + \omega^2} = \lim_{\alpha \rightarrow 0} \left[\arctan \frac{\omega}{\alpha} \right]_{\omega \rightarrow -\infty}^{\omega \rightarrow +\infty} = \pi$$

On the other hand, we have

$$\lim_{\alpha \rightarrow 0} \int_{-\infty}^{+\infty} \frac{\alpha d\omega}{\alpha^2 + \omega^2} = \int_{-\infty}^{+\infty} c\delta(\omega) d\omega = c$$

Relating the two, we get $c = \pi$. Therefore, the Fourier Transform of the step function becomes

$$F(\omega) = \pi\delta(\omega) - \frac{i}{\omega}$$

(ii) The power spectral distribution function is $|F(\omega)|^2$, so it has a power law behavior $1/\omega^2$ which is similar to the spectrum of Brown noise as discussed in the lecture.

(d) Sign Function: $f(t) = \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$

Same limiting method ($e^{-\alpha t}$ for $t > 0$, $e^{\alpha t}$ for $t < 0$):

$$F(\omega) = \lim_{\alpha \rightarrow 0} \left(\frac{1}{\alpha + i\omega} - \frac{1}{\alpha - i\omega} \right) = \lim_{\alpha \rightarrow 0} \frac{-2i\omega}{\alpha^2 + \omega^2}$$

$$\boxed{F(\omega) = \frac{-2i}{\omega}}$$

(No delta at origin since the contributions cancel.)

(e) Using Duality to Find Further FTs:

Key duality idea: Inverse FT is $f(t) = \frac{1}{2\pi} \int F(\omega) e^{i\omega t} d\omega$.

Redefine $t \rightarrow -\omega$, $\omega \rightarrow t'$ to swap the roles of time and frequency.

(i) Instead of directly calculating the Fourier transform of $f(t) = 1/t$ which involves contour integrals, we will use the duality property here. Recall that the inverse Fourier transform of a function is defined as

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

For the step function in (d), we can get it back by applying the inverse Fourier transform on $-2i/\omega$.

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{-2i}{\omega} \right) e^{i\omega t} d\omega = \begin{cases} 1 & (t > 0) \\ -1 & (t < 0) \end{cases}$$

Redefining the parameters in the integral, i.e. $t \rightarrow -\omega$ and $\omega \rightarrow t'$, we get

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{-2i}{\omega} \right) e^{i\omega t} d\omega &= \frac{-i}{\pi} \int_{-\infty}^{\infty} \left(\frac{1}{t'} \right) e^{-i\omega t'} dt' \\ &= \begin{cases} 1 & (-\omega > 0) \\ -1 & (-\omega < 0) \end{cases} = \begin{cases} -1 & (\omega > 0) \\ 1 & (\omega < 0) \end{cases} = -\text{sign}(\omega) \end{aligned}$$

Therefore, the Fourier Transform of $f(t) = 1/t$ is

$$F(\omega) = \int_{-\infty}^{\infty} \left(\frac{1}{t'} \right) e^{-i\omega t'} dt' = \frac{\pi}{i} \text{sign}(\omega) = -i\pi \text{sign}(\omega)$$

(ii) Using the result in (d), the Fourier Transform of $f(t) = |t|$ is

$$\begin{aligned} F(\omega) &= \lim_{\alpha \rightarrow 0} \left(\int_0^{\infty} t e^{-\alpha t} e^{-i\omega t} dt + \int_{-\infty}^0 t e^{\alpha t} e^{-i\omega t} dt \right) \\ &= \lim_{\alpha \rightarrow 0} \left(-\frac{d}{d\alpha} \int_0^{\infty} e^{-\alpha t} e^{-i\omega t} dt - \frac{d}{d\alpha} \int_{-\infty}^0 e^{\alpha t} e^{-i\omega t} dt \right) \\ &= \lim_{\alpha \rightarrow 0} \left[-\frac{d}{d\alpha} \left(\frac{1}{\alpha + i\omega} + \frac{1}{\alpha - i\omega} \right) \right] \\ &= \lim_{\alpha \rightarrow 0} \left[-\frac{d}{d\alpha} \left(\frac{2\alpha}{\alpha^2 + \omega^2} \right) \right] \\ &= \lim_{\alpha \rightarrow 0} \left[-\frac{2(\alpha^2 + \omega^2) + (2\alpha)^2}{(\alpha^2 + \omega^2)^2} \right] = \frac{-2}{\omega^2} \end{aligned}$$

(iii) Using the Fourier transform of $|t|$ and the duality property again, we get $F(\omega) = -\pi|\omega|$ for the Fourier transform of $f(t) = 1/t^2$.

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{-2}{\omega^2} \right) e^{i\omega t} d\omega = |t|$$

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \left(\frac{-1}{t'^2} \right) e^{-i\omega t'} dt' = -|\omega| = |\omega|$$

$$F(\omega) = \int_{-\infty}^{\infty} \left(\frac{1}{t'^2} \right) e^{-i\omega t'} dt' = -\pi|\omega|$$